

Assignment: AI System Specification

Roll No: 2K23/CSE/106

Submitted To: Sir Rajesh Kumar

Task 1: PEAS Specification

The system proposed is an **AI Tutor for Personalized Student Learning Recommendations**. Its core objective is to adaptively guide students through educational material based on their individual learning pace, comprehension levels, and engagement history.

Performance Measures

- Improvement in student test and quiz scores over the duration of the semester.
- Increase in the total time students actively engage with recommended materials.
- Reduction in dropout or failure rates for traditionally difficult modules.
- High user satisfaction ratings gathered through periodic student feedback surveys.

Environment

- Digital Learning Management Systems (LMS) such as Canvas, Blackboard, or Moodle.
- Student hardware devices (laptops, tablets, and smartphones) displaying the interface.
- Structured databases containing course modules, video lectures, and question banks.

Actuators

- User interface (UI) displays that highlight dynamically generated course paths.
- Automated email and push notifications for study reminders and achievement milestones.
- Dynamic quiz generators that adjust question difficulty on the fly.
- Progress dashboard visualizers displaying the student's real-time mastery of topics.

Sensors

- Navigation trackers capturing clicks, mouse movements, and platform interactions.
- Video analytics monitoring watch-time, pauses, skips, and rewinds.
- Quiz grading modules that record correct answers, incorrect attempts, and time taken per question.
- Time-on-page analytics and textual input fields for user search queries.

Task 2: Environment Classification

- **Partially Observable:** The AI cannot access the student's internal mental state, off-screen study habits, or physical distractions, relying entirely on digital interaction proxies.
- **Multi-agent:** The system operates within an environment populated by human agents (students and human instructors) and potentially other educational software agents.
- **Stochastic:** The outcome of recommending a specific learning resource is highly unpredictable because human cognitive load and motivation fluctuate constantly.
- **Sequential:** A recommendation made today directly alters the student's knowledge base, fundamentally shifting the baseline for all future recommendations.
- **Dynamic:** The environment continuously changes as the student's comprehension evolves and as instructors upload new learning materials or modify curriculum formats.
- **Continuous:** Time spent learning, performance grading scales, and the influx of behavioral data occur over a continuous spectrum rather than in discrete, isolated turns.

Task 3: Critical Analysis

1. System Architecture Justification

A utility-based architecture is fundamentally more appropriate for an AI tutor than a simple goal-based architecture because educational success is a spectrum, not a binary state. A standard goal-based agent might blindly aim to help a student pass a quiz, potentially recommending the easiest path or encouraging rote memorization over deep understanding. In contrast, a utility function balances multiple overlapping variables—such as long-term knowledge retention, active engagement, and the appropriate level of challenge. By maximizing overall utility, the system calculates the optimal trade-off between pushing the student out of their comfort zone and preventing burnout, resulting in a higher quality, highly personalized learning experience.

2. Ethical and Privacy Considerations

Deploying an AI tutor necessitates the continuous tracking of student behaviors, which raises significant data privacy and surveillance concerns. There is a persistent risk that algorithmic bias could unfairly profile students—for instance, consistently recommending lower-tier material to specific users based on flawed historic training data, thereby limiting their academic potential. Mitigating these risks requires implementing strict data anonymization protocols to protect identities, ensuring the recommendation algorithms are transparent, and maintaining robust human-in-the-loop oversight so that human educators can review and override misaligned AI assessments.

Structured Representation

```
{
  "system_name": "AI Tutor for Personalized Student Learning
Recommendations",
  "peas": {
    "performance_measures": [
      "Improvement in student test scores",
      "Increase in active engagement time with materials",
      "Reduction in module failure rates",
      "Student feedback satisfaction ratings"
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    "environment": [
      "Digital learning management systems (LMS)",
      "Student hardware devices (laptops, tablets)",
      "Databases of educational content"
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    "actuators": [
      "UI recommendation displays",
      "Automated study notifications",
      "Dynamic adaptive quiz generators",
      "Interactive progress dashboards"
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    "sensors": [
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      "Quiz grading modules",
      "Time-on-page analytics"
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    "agent_type": "Multi-agent",
    "determinism": "Stochastic",
    "episodicity": "Sequential",
    "dynamism": "Dynamic",
    "dimensionality": "Continuous"
  },
  "utility_function": "Maximize(Learning_Gain * Engagement_Score) -
Minimize(Student_Frustration)"
}
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